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EDITORIAL

It is with great pleasure and a deep sense of academic fulfillment that we present the maiden edition of the Public Sector Accounting and Finance Journal (PSAFJ), published by ANAN University, Kwall. This inaugural issue marks a significant milestone in the University's commitment to advancing scholarship, policy, and practice in the field of public sector accounting, finance, and governance.

In an era when nations are striving for transparency, accountability, and fiscal sustainability, the role of public sector accounting and finance has never been more critical. The journal is therefore conceived as a platform for intellectual engagement, rigorous research, and policy discourse that contribute meaningfully to the strengthening of public financial management systems, good governance, and sustainable development in Nigeria and beyond.

This maiden edition features ten scholarly articles drawn from a wide spectrum of research interests and methodologies within the field. The papers collectively address critical issues ranging from public sector financial reporting and accountability, budgeting reforms, internal control mechanisms, audit quality, fiscal transparency, to the application of emerging technologies in public finance management. Each contribution reflects the authors' depth of research and commitment to the continuous improvement of public sector governance and accountability.

COLLEGE OF PUBLIC SECTOR ACCOUNTING
ANUK Public Sector Accounting and Finance Journal (APSAFJ)

THE OBJECTIVE OF THE JOURNAL

To provide a platform for researchers and academics to publish research papers and articles within the research domains of public sector accounting, public finance, public economics, environmental, oil and gas accounting, public financial management, taxation and fiscal policy, public policy and finance, macroeconomics and development, public treasury management, and other related areas..

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2. Statement of Research Problem

3. Research Hypotheses or Objectives

*Review of Related Literature

4. Data Presentation & Analysis

5. Research Findings & Discussion of Findings

6. Conclusion, Recommendation & Policy Implication

7. References

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EFFECT OF TETFUND INTERVENTION ON RESEARCH AND DEVELOPMENT IN NIGERIAN TERTIARY INSTITUTIONS

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Abstract

This study examines the effect of TETFUND interventions on research and development (R&D) in Nigerian tertiary institutions, focusing on research grant allocations, academic staff participation in TETFUND-funded training, and the availability of research infrastructure. Using a sample of 160 academic staff across universities, polytechnics, and colleges of education, data were analyzed using descriptive statistics, correlation, and multiple linear regression. The results reveal that research grant allocation and infrastructure availability have significant positive effects on research output, while participation in TETFUND-funded training does not significantly influence research productivity. The study recommends that TETFUND and institutional administrators should ensure that research grant application processes are transparent, timely, and inclusive to allow broader participation among academic staff. The findings support prior studies emphasizing the role of funding and infrastructure in enhancing research capacity, while highlighting the need for redesigning training programs to yield measurable outcomes. The study provides evidence-based recommendations for policy and institutional practice, contributing to the optimization of research interventions in Nigerian higher education.

Keywords: Research & development, research capacity, institutional practice, intervention.

1. INTRODUCTION

Research and development (R&D) have become critical drivers of economic growth, technological advancement, and societal progress across the globe. Developed countries such as the United States, China, and members of the European Union have consistently demonstrated strong commitment to R&D by allocating a significant proportion of their GDP to innovation and scientific inquiry. These investments have yielded significant outcomes, including high publication outputs, patents, and knowledge-based industries. The Organisation for Economic Co-operation and Development (OECD) countries, for instance, invest on average about 2.4% of their GDP in R&D, with countries like South Korea and Israel exceeding 4%. These figures illustrate the growing recognition of R&D as a strategic priority and a foundation for long-term competitiveness. In advanced economies, tertiary institutions play a vital role in generating research outputs that feed into national innovation systems. Their capacity to conduct cutting-edge research is sustained by deliberate policies, substantial funding, and structured development programs targeting faculty and infrastructure. Consequently, higher

education institutions in developed countries are not only centers of learning but also pivotal engines of research-driven development.

In contrast, R&D in Africa remains underfunded and fragmented. The continent accounts for less than 1% of global research output despite having 12% of the world's population. African countries generally invest less than 0.5% of their GDP in R&D, falling short of the African Union's recommendation of 1%. Nigeria, the largest economy in Africa, is no exception. Its tertiary institutions face challenges such as poor funding, inadequate infrastructure, low staff morale, and weak research capacity. These limitations have impeded the nation's ability to produce significant scientific and technological breakthroughs. In response to these systemic challenges, the Nigerian government established the Tertiary Education Trust Fund (TETFUND) to provide targeted financial interventions aimed at revitalizing the research culture within tertiary institutions. Through annual disbursements, TETFUND supports academic staff training, research grants, infrastructure development, and conference participation. While the establishment of TETFUND has been a commendable step towards improving R&D in

Nigeria, the extent to which these interventions translate into tangible research outputs remains an empirical question. Some studies have reported positive impacts on staff development and infrastructural improvement (Bello et al., 2024; Anaelobi & Agim, 2019), but there is limited consensus on the overall effectiveness of these interventions across different institutions and regions.

This study investigates the effect of TETFUND interventions on research and development in Nigerian tertiary institutions, with a specific focus on three key independent variables: research grant allocations, academic staff participation in TETFUND-funded training, and the availability of TETFUND-supported research infrastructure. Research grant allocation refers to the monetary support provided to institutions or individual researchers for conducting research projects. The underlying assumption is that increased funding enhances research productivity by enabling scholars to procure materials, access databases, and disseminate findings. Academic staff participation in TETFUND-sponsored training, such as postgraduate studies, workshops, and international conferences, is expected to build research capacity and improve scholarly output. Similarly, the availability of research infrastructure—labs, equipment, and ICT resources—funded by TETFUND is anticipated to create an enabling environment for research activities. The dependent variable, research and development, is measured by the quantity and quality of research outputs such as peer-reviewed publications, patents, and completed projects. The conceptual linkage between these variables is grounded in the idea that capacity-building interventions, when adequately funded and strategically implemented, directly enhance institutional research performance.

The extant literature on TETFUND's impact provides valuable insights but also reveals several limitations. For example, Oguejiofor, Nzeribe, and Ugwuogo (2024a) reported that TETFUND-funded training had a statistically significant effect on staff retention and quality of education in public universities in Southeast Nigeria. Similarly, Bello et al. (2024) used regression analysis to establish a modest but positive relationship between research grant funding and staff development at the Federal University Dutsinma. Aminu and Aminu (2018), focusing on Ramat Polytechnic Maiduguri, found a notable increase in academic staff publication volume linked to TETFUND support. However, many of these studies are geographically limited and institution-specific,

focusing either on federal universities or polytechnics in particular regions. Moreover, most employ descriptive survey designs and basic statistical tests like t-tests or chi-square analyses, which, while useful, may not capture the complex, multi-variable relationships influencing research performance. Anaelobi and Agim (2019) observed infrastructural improvements in Southeast university libraries due to TETFUND, but did not assess how such improvements impacted research outcomes. In sum, although there is evidence that TETFUND interventions contribute to improving tertiary education in Nigeria, the literature lacks comprehensive, multi-variable studies that link funding, training, and infrastructure directly to research and development outcomes across a wider sample of institutions.

In light of the literature review, this study identifies several critical gaps. Practically, there is limited understanding of how different types of TETFUND interventions interact to influence research productivity. Most institutions receive funds for infrastructure, training, and research simultaneously, yet little is known about their combined effects. Additionally, geographic disparities in the allocation and utilization of TETFUND resources remain underexplored. Methodologically, the reliance on cross-sectional surveys and limited statistical tools restricts the depth of inference that can be drawn from existing studies. Few studies employ rigorous quantitative techniques capable of isolating the independent and joint effects of TETFUND interventions. There is also a literature gap in the integrated assessment of all three major components of TETFUND—funding, training, and infrastructure—on R&D outcomes. Most studies treat these variables in isolation rather than examining their synergistic effects. This study addresses these gaps by employing a multi-variable framework to evaluate the direct and joint effects of TETFUND research grants, training participation, and infrastructure availability on research and development outcomes in Nigerian tertiary institutions.

Based on the above analysis, the specific objectives of this study are:

- i. To examine the effect of TETFUND research grant allocations on research and development in Nigerian tertiary institutions.
- ii. To investigate how academic staff participation in TETFUND-funded training programs influences research and development outcomes.

- iii. To assess the impact of the availability of TETFUND-supported research infrastructure on research and development in Nigerian tertiary institutions.

In line with the objectives, the following null hypotheses were stated:

H₀₁: TETFUND research grant allocations have no significant effect on research and development in Nigerian tertiary institutions.

H₀₂: Academic staff participation in TETFUND-funded training programs has no significant effect on research and development in Nigerian tertiary institutions.

H₀₃: The availability of TETFUND-supported research infrastructure has no significant effect on research and development in Nigerian tertiary institutions.

The scope of the study includes federal and state-owned universities, polytechnics, and colleges of education across Nigeria that have benefited from TETFUND interventions between 2018 and 2024. Data will be collected through institutional records, survey questionnaires, and structured interviews targeting academic staff and administrative officers responsible for TETFUND project implementation. The study's significance lies in its potential to inform evidence-based policy decisions and improve the strategic implementation of TETFUND interventions. For policymakers, the findings will offer insights into which areas of investment yield the greatest returns in terms of research output. For tertiary institutions, the results can guide internal planning, resource allocation, and institutional capacity-building efforts. Methodologically, this research contributes to literature by adopting a multi-dimensional and data-driven approach to evaluating public sector interventions in higher education. In sum, the study is positioned to advance both theoretical understanding and practical management of research and development in Nigerian tertiary education.

2. LITERATURE REVIEW

2.1 Conceptual Review

2.1.1 Research and Development

Research and development (R&D) in tertiary education represent the systematic effort by academic institutions to generate new knowledge, innovate, and provide evidence-based solutions

to societal problems. In developed countries, R&D forms the foundation of technological advancement, driving patent generation, industrial growth, and human capital development. In contrast, Nigerian tertiary institutions have historically struggled to sustain meaningful research activities due to low funding, poor infrastructure, and weak institutional support (Bello et al., 2024). However, interventions such as those provided by the Tertiary Education Trust Fund (TETFUND) aim to strengthen the research ecosystem by supporting academic staff and institutions in enhancing their research outputs (Oguejiofor et al., 2024b). Effective R&D in higher education is closely linked to faculty development, research funding, and access to enabling infrastructure, all of which are crucial for increasing scholarly productivity and the global competitiveness of Nigerian universities and polytechnics.

2.1.2 Research Grant Allocations

TETFUND research grant allocations are financial resources disbursed to tertiary institutions and their academic staff to facilitate research projects, publications, and innovative development. These grants are crucial for overcoming one of the most significant barriers to research productivity in Nigeria: inadequate funding. Grant allocations support fieldwork, data collection, access to journals, conference participation, and the publication of research findings. Bello et al. (2024) found that TETFUND-sponsored research grants contributed significantly to staff development and improved research capacity at the Federal University Dutsinma. Similarly, Aminu and Aminu (2018) observed a direct relationship between research grant accessibility and publication output among staff at Ramat Polytechnic. The strategic allocation of research funding is, therefore, an essential driver of scholarly output and a catalyst for institutional research excellence.

2.1.3 Academic Staff Participation in TETFUND-funded Training

Academic staff participation in TETFUND-funded training encompasses involvement in sponsored programs such as postgraduate studies, research workshops, academic conferences, and international fellowships. These interventions are designed to build research competencies, enhance teaching effectiveness, and expose staff to global academic standards. Evidence suggests that staff who undergo TETFUND-sponsored training are more productive in terms of research

output and quality (Oguejiofor et al., 2024a). Participation in such training initiatives also improves retention and motivation, as it creates clear career development pathways and opportunities for scholarly networking. By investing in human capital, TETFUND contributes not only to individual academic growth but also to institutional research advancement. The training variable thus plays a mediating role in translating funding and policy into tangible R&D outcomes within Nigerian tertiary institutions.

2.1.4 Availability of TETFUND-Supported Research Infrastructure

The availability of TETFUND-supported research infrastructure refers to the presence and accessibility of laboratories, ICT tools, equipment, and specialized facilities funded through TETFUND interventions. Infrastructure is foundational to meaningful research activity, particularly in science, engineering, and applied social sciences where empirical investigation and data analysis require reliable tools and environments. Anaelobi and Agim (2019) reported that TETFUND interventions significantly improved the state of university libraries and laboratories in Southeast Nigeria, creating more conducive environments for research. When infrastructure is readily available and functional, academic staff are better positioned to execute funded research projects, attract collaborations, and produce high-impact studies. Therefore, institutional access to modern research infrastructure is not just an operational necessity but a determinant of research quality and innovation potential.

2.2 Empirical Literature Review

Empirical investigations into the role of TETFUND in enhancing research and development (R&D) in Nigerian tertiary institutions consistently reveal that financial interventions through research grants significantly influence academic productivity. Bello et al. (2024) conducted a quantitative study at Federal University Dutsinma and found that TETFUND-sponsored research grants positively impacted staff development and modestly increased research output. Similarly, Aminu and Aminu (2018) observed that access to research funds correlated with an increase in the volume of publications among staff at Ramat Polytechnic Maiduguri. Supporting this view, Nwakanma (2025) reported that TETFUND funding significantly contributed to infrastructural and research development in selected public institutions. These findings

suggest that research grant allocations not only facilitate access to materials and data but also enable scholars to disseminate their findings more widely. However, some studies note that despite financial support, administrative bottlenecks and delayed fund disbursement can undermine the effectiveness of these grants.

A related line of inquiry focuses on the influence of academic staff participation in TETFUND-funded training on research performance. Oguejiofor, Nzeribe, and Ugwuogo (2024a), using a descriptive survey of 336 lecturers across Southeast Nigerian universities, found that participation in TETFUND-sponsored training significantly improved both research output and staff retention. Their follow-up study (2024b) confirmed that training and capacity-building initiatives had a positive impact on the quality of education and research productivity. In the same vein, Olamide and Olaniran (2024) discovered that academic staff who participated in conferences and postgraduate training through TETFUND were more likely to engage in collaborative research and publish in high-impact journals. These findings demonstrate that investments in human capital development are a key mechanism through which TETFUND contributes to research advancement. However, some researchers argue that the uneven distribution of training opportunities across institutions limits the broader impact of such interventions.

Infrastructural development, as another dimension of TETFUND's mandate, also plays a critical role in shaping research output. Anaelobi and Agim (2019) assessed the effect of TETFUND-supported library and laboratory upgrades in Southeast universities and concluded that improved infrastructure created enabling environments for academic research. Ogunode and Agbor (2024) further emphasized that infrastructure—particularly in guidance and counselling departments—enhanced the capacity of researchers to conduct fieldwork and data analysis. Similarly, Aprebo and Onyeike (2018) highlighted that the availability of improved research spaces such as laboratories and study centers contributed to increased research engagement among faculty in Rivers and Bayelsa States. Despite these positive results, Bamigboye et al. (2015) cautioned that poor maintenance of TETFUND-built infrastructure sometimes leads to rapid deterioration, thereby limiting long-term benefits. These studies collectively underscore that while physical infrastructure is vital to R&D performance, sustainability and institutional commitment to facility upkeep are also essential.

Other studies have attempted to evaluate the combined or overall effect of TETFUND interventions on institutional research capacity. For instance, Oguejiofor (2022) analyzed the relationship between administrative support, academic training, and infrastructure development in federal universities and found a significant cumulative effect of these variables on research productivity. Likewise, Aminu et al. (2021) conducted a chi-square-based analysis in Northwestern Nigeria and confirmed that institutions receiving sustained TETFUND support showed higher growth in academic output compared to those with less consistent funding. Meanwhile, Ezeali (2017) found that human resource development driven by TETFUND funding directly contributed to a rise in postgraduate research activity. While these studies affirm the positive influence of TETFUND interventions, they also highlight the need for improved policy coordination, transparency in fund allocation, and equitable distribution of resources across geopolitical zones. Overall, the literature points to the critical role of TETFUND in revitalizing research functions in Nigerian tertiary institutions, though further empirical refinement is needed to assess long-term outcomes and institutional differences.

2.3 Theoretical Framework

This study is anchored on two key theories: Human Capital Theory and Resource-Based View (RBV). The Human Capital Theory, originally advanced by Schultz (1961) and later refined by Becker (1993), posits that investment in people through education, training, and capacity building leads to increased productivity and innovation. Within the context of this research, the theory provides a foundation for understanding the link between academic staff participation in TETFUND-funded training and improvements in research and development outcomes. By funding postgraduate education, research workshops, and conference attendance, TETFUND enhances the skills, knowledge, and motivation of academic personnel, thereby increasing their research output and academic engagement (Oguejiofor et al., 2024a; Olamide & Olaniran, 2024). The theory implies that tertiary institutions with better-trained staff are more likely to generate impactful research, secure external grants, and contribute to national innovation goals.

The Resource-Based View (RBV), introduced by Barney (1991), complements this perspective by emphasizing the strategic value of internal resources in gaining competitive advantage. According to RBV, institutions that possess rare,

valuable, and well-organized resources are better positioned to achieve superior performance outcomes. In the context of Nigerian tertiary institutions, TETFUND interventions serve as strategic resources that enhance research capacity. Access to research funding, laboratory facilities, and digital infrastructure (Anaelobi & Agim, 2019; Ogunode & Agbor, 2024) enables institutions to conduct high-quality research, attract collaborations, and compete globally. Thus, RBV supports the argument that the availability and effective utilization of TETFUND-supported resources can significantly influence research and development outputs. Together, Human Capital Theory and RBV provide a robust theoretical lens through which the impact of TETFUND interventions on R&D in Nigerian tertiary institutions can be analyzed.

3. METHODOLOGY

This study adopted a quantitative research design, specifically the survey research method, to examine the effect of TETFUND interventions on research and development in Nigerian tertiary institutions. The choice of survey design is informed by its suitability for collecting standardized data from a large population, enabling statistical generalization and comparison across variables. This approach is consistent with prior studies such as those by Bello et al. (2024) and Oguejiofor et al. (2024a), who employed survey designs to explore the influence of TETFUND grants and training on staff development and institutional research output. The study is guided by a positivist research philosophy, which assumes that reality is objective and measurable, and that knowledge can be obtained through empirical observation and statistical analysis. This paradigm aligns with the use of structured questionnaires, hypothesis testing, and the pursuit of generalizable findings, typical of research seeking to quantify relationships between institutional interventions and research productivity.

The population of the study comprised academic staff from selected federal and state universities, polytechnics, and colleges of education in Nigeria that received TETFUND support between 2018 and 2024. A multistage sampling technique was used. First, six institutions were purposively selected across Nigeria's geopolitical zones to ensure representation. Then, stratified random sampling was applied to select respondents across faculties or departments, resulting in a final sample of 160 academic staff. The study relied on primary data, collected using

a structured questionnaire adapted from prior empirical studies on TETFUND’s impact (e.g., Oguejiofor et al., 2024b; Aminu & Aminu, 2018). The questionnaire consisted of five sections capturing demographic details, access to research grants, participation in TETFUND training, availability of infrastructure, and research output. Validity was ensured through expert review, while reliability was confirmed using Cronbach’s alpha, which yielded acceptable values above 0.70.

Data collected were analyzed using descriptive statistics (mean, frequency, standard deviation) and inferential statistics, specifically multiple linear regression analysis, to examine the relationship between the independent variables and the dependent variable. This analytical approach allowed the study to determine both the individual and combined effects of TETFUND interventions on research and development outcomes, consistent with the analytical methods adopted by Bello et al. (2024) and Aminu et al. (2021). The regression model is stated as follows:

$$ROP_i = \alpha_i + \beta_1RGA_i + \beta_2PTT_i + \beta_3INF_i + \varepsilon_i$$

Where:

ROP = Research Output

RGA = Research Grant Allocation

PTT = Academic Staff Participation

INF = Availability of Research Infrastructure

The variables were measured as follows:
Research and Development (dependent variable)

was measured by the number of peer-reviewed publications, completed research projects, and patents reported in the last three years. The independent variables were measured as follows: Research Grant Allocation by the amount and frequency of TETFUND-funded projects accessed; Academic Staff Participation in TETFUND-funded Training by the number of conferences, workshops, and postgraduate programs sponsored by TETFUND; and Availability of Research Infrastructure by the presence and adequacy of TETFUND-supported laboratories, libraries, and ICT tools (Anaelobi & Agim, 2019; Oguejiofor, 2022). Statistical analysis was conducted using Stata (version 13), and results were interpreted at a 5% level of significance.

4. RESULTS AND DISCUSSION

This section presents and discusses the results of the empirical analysis conducted to examine the effect of TETFUND interventions on research and development in Nigerian tertiary institutions. The analysis draws on data obtained from 160 academic staff and is structured around the key variables identified in the study: research grant allocations (RGA), academic staff participation in TETFUND-funded training (PTT), and availability of research infrastructure (INF). Descriptive statistics, correlation analysis, and multiple regression techniques were employed to evaluate the individual and combined effects of these variables on research output (ROP). The results are interpreted in light of the research hypotheses, existing literature, and theoretical foundations such as Human Capital Theory and the Resource-Based View. Emphasis is placed not only on statistical significance but also on the practical implications of the findings for policy and institutional strategy.

Table 1 Descriptive Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
ROP	160	7.0625	2.836615	1	14
RGA	160	3.1	1.501362	0	5

PTT	160	4.6125	3.252454	0	10
INF	160	1.6625	.800059	1	3

Table 1 presents the descriptive statistics for the four main variables in the study: research output (ROP), research grant allocation (RGA), academic staff participation in TETFUND-funded training (PTT), and availability of research infrastructure (INF). The results are based on data from 160 respondents. The dependent variable, research output (ROP), has a mean of 7.06 with a standard deviation of 2.84, indicating that, on average, academic staff produced approximately seven research outputs over the past three years, with moderate variation across respondents. The minimum value is 1 and the maximum is 14, suggesting that while some respondents had very low output, others were relatively productive in their research activities.

The first independent variable, research grant allocation (RGA), shows a mean of 3.1, with a standard deviation of 1.50, ranging from 0 to 5. This implies that respondents had access to about three TETFUND-supported research projects on average, though some had none, indicating variation in access to research funding.

Participation in TETFUND-funded training (PTT) has a mean value of 4.61 and a relatively high standard deviation of 3.25, with values ranging from 0 to 10. This wide spread suggests considerable differences in the extent of staff participation in TETFUND-sponsored conferences, workshops, or postgraduate programs. While some academic staff may not have participated at all, others had access to up to 10 training events. Lastly, the availability of TETFUND-supported infrastructure (INF) has a mean of 1.66 and a standard deviation of 0.80. Since the variable is measured on an ordinal scale (1 = High, 2 = Moderate, 3 = Low), the average value near 2 suggests that most respondents rated the availability of infrastructure as moderate to high. The spread of responses across the three categories highlights differences in institutional infrastructure, which may affect research capacity.

The normality test using the Shapiro-Wilk statistic assessed whether the variables in the study followed a normal distribution.

Table 2 Normality Test (Shapiro-Wilk)

Variable	Obs	W	V	z	Prob>z
ROP	160	0.99454	0.672	-0.904	0.81705
RGA	160	0.98401	1.967	1.538	0.06197
PTT	160	0.97886	2.600	2.173	0.01488
INF	160	0.98450	1.906	1.468	0.07111

As shown in Table 2, the probability values (p-values) for ROP ($p = 0.81705$), RGA ($p = 0.06197$), and INF ($p = 0.07111$) are all greater than the 0.05 threshold, indicating that these variables do not significantly deviate from normality and can be considered approximately normally distributed. However, PTT ($p = 0.01488$) has a p-value less than 0.05, suggesting a statistically significant deviation from normality. Despite this, the overall data structure remains suitable for parametric analysis given the large sample size ($n = 160$), and multiple

regression techniques are still considered robust in the presence of slight non-normality in one of the predictors.

The correlation analysis presented in Table 3 shows the degree and direction of linear relationships between the dependent variable, research output (ROP), and the independent variables: research grant allocation (RGA), participation in TETFUND-funded training (PTT), and availability of research infrastructure (INF).

Table 3 *Correlation Analysis*

Variables	ROP	RGA	PTT	INF
ROP	1.0000			
RGA	0.4253	1.0000		
PTT	0.0579	0.0325	1.0000	
INF	0.1867	0.0387	0.1355	1.000

The results reveal a moderate positive correlation between ROP and RGA ($r = 0.4253$), suggesting that academic staff with more access to research grants tend to have higher research output. On the other hand, the correlation between ROP and PTT is weak and positive ($r = 0.0579$), indicating a minimal linear relationship between participation in TETFUND-sponsored training programs and research output. Similarly, the correlation between ROP and INF is low but positive ($r = 0.1867$), implying that availability of infrastructure has a slight association with increased research productivity. Although these relationships are weaker than that between ROP and RGA, their positive directions suggest that training and infrastructure may still play complementary roles in enhancing research outcomes. Additionally, the low correlations among the independent variables (e.g., RGA and PTT, $r = 0.0325$) indicate minimal multicollinearity, which supports the reliability

of including all three predictors in the regression model.

The multiple-linear regression results in Table 4 evaluate how each TETFUND intervention predicts research output (ROP) while controlling for the other interventions. The overall model is statistically significant ($F = 13.06, p < 0.001$), and the R-squared of 0.2104 indicates that about 21 percent of the variance in research productivity across the 160 academics is explained jointly by research-grant allocations (RGA), participation in TETFUND-funded training (PTT), and the availability of research infrastructure (INF).

The multiple-linear regression results in Table 4 evaluate how each TETFUND intervention predicts research output (ROP) while controlling for the other interventions.

Table 4 *Regression Analysis*

Variables	Coef.	Std. Err.	t	P>t
RGA	.789	.145	5.45	0.000
PTT	.019	.067	0.28	0.778
INF	.594	.261	2.27	0.021
CONS	3.539	.614	5.77	0.000
R-Squared		0.2104		
F		13.06		
Probability		0.000		

The overall model is statistically significant ($F = 13.06, p < 0.001$), and the R-squared of 0.2104 indicates that about 21 percent of the variance in research productivity across the 160 academics is explained jointly by research-grant allocations (RGA), participation in TETFUND-funded training (PTT), and the availability of research infrastructure (INF). Variance-inflation factors averaging 1.01 confirm that multicollinearity is negligible, so the coefficient estimates are stable and interpretable.

Research-grant allocation (RGA) has a positive, sizable, and highly significant coefficient ($\beta = 0.789, t = 5.45, p < 0.001$). *Ceteris paribus*, every additional TETFUND-supported grant is associated with roughly 0.79 extra research outputs over the three-year window, underscoring the importance of direct funding in stimulating scholarly productivity—an effect consistent with Bello et al. (2024) and Aminu & Aminu (2018). Infrastructure availability (INF) also exhibits a positive and

statistically significant relationship with research output ($\beta = 0.594, t = 2.27, p = 0.021$). Since INF is coded 1 = High, 2 = Moderate, 3 = Low, a one-unit improvement (e.g., from “moderate” to “high”) raises expected research output by nearly 0.6 publications, echoing the infrastructure findings of Anaelobi & Agim (2019). In contrast, participation in TETFUND-funded training (PTT) shows a very small, non-significant coefficient ($\beta = 0.019, t = 0.28, p = 0.778$). The data therefore provide no evidence that additional training events—at least as captured here—translate into higher short-term publication counts, even though previous survey studies suggested qualitative benefits (Oguejiofor et al., 2024a). The significant constant (3.539, $p < 0.001$) represents the baseline research output for staff who received no grants, attended no training, and rated their infrastructure as “low.”

Table 5 presents the test of hypotheses.

Table 5 Hypotheses Testing

Null hypothesis	Decision	Evidence
H₀₁: RGA has no significant effect on R&D. Reject H₀₁		$p < 0.001$ for RGA
H₀₂: PTT has no significant effect on R&D. Fail to reject H₀₂		$p = 0.778$ for PTT
H₀₃: INF has no significant effect on R&D. Reject H₀₃		$p = 0.021$ for INF
The evidence confirms that TETFUND research grants and improved research infrastructure exert meaningful, positive influences on academic research productivity, whereas staff participation in TETFUND training does not produce an immediate, measurable boost in publication output. The joint significance of the model underscores TETFUND’s overall relevance, but it also signals that channeling resources toward grant funding and infrastructure may yield the most tangible short-term gains in research and development across Nigerian tertiary institutions. The results of the regression analysis provide compelling evidence that TETFUND research grant allocations significantly enhance research and development in Nigerian tertiary institutions. The strong positive relationship between research grant access and research output aligns with the findings of Bello et al. (2024), who reported that TETFUND-funded projects contributed to improved staff development and publication output at Federal University Dutsinma. Similarly, Aminu and Aminu (2018) found that the availability of TETFUND grants significantly influenced the volume of research publications among academic staff in Ramat Polytechnic Maiduguri. These findings suggest that access to direct financial support empowers researchers to execute projects more efficiently, access relevant materials, and disseminate findings, thereby improving productivity. The consistency of this result across different institutional settings reinforces the argument that financial investment in research is one of the most effective levers for stimulating innovation in the higher education sector. In contrast, the analysis revealed that academic staff participation in TETFUND-funded training had no statistically significant effect on research output, a result that stands in partial contrast to some earlier studies. For example, Oguejiofor, Nzeribe, and Ugwuogo (2024a, 2024b) reported that training programs significantly enhanced academic staff performance and research engagement in Southeast Nigerian universities. While their studies were based on descriptive survey designs and subjective assessments, the current study’s regression model shows that increased participation in training alone may not translate into immediate publication output. One possible explanation is that the benefits of training programs—such as skill acquisition and exposure—may take longer to manifest in measurable research output or may enhance research quality rather than quantity. Furthermore, variation in the quality or relevance of training programs across institutions could dilute their overall effect. Thus, while training remains important for capacity building, it may not yield immediate results in terms of publication count or completed research projects. The third major finding was the significant positive relationship between availability of TETFUND-supported research infrastructure and research output. This supports earlier findings by Anaelobi and Agim (2019), who noted that infrastructural improvements—especially in university libraries and laboratories—provided more conducive environments for scholarly activities. Likewise, Ogunode and Agbor (2024) emphasized the role of functional infrastructure in enhancing research efficiency and data analysis capabilities. The present study’s results confirm that modern facilities funded by TETFUND play a critical role in creating the physical conditions necessary for academic staff to conduct research. This is particularly important in applied fields like engineering, agriculture, and the sciences, where quality infrastructure directly influences the feasibility and scope of empirical work. Taken together, the findings suggest that TETFUND’s dual focus on research grants and infrastructure has a stronger and more immediate impact on		

research and development than its training programs. The study thus offers evidence-based guidance for policy makers and university administrators seeking to optimize TETFUND interventions for maximum research productivity.

5. CONCLUSION AND RECOMMENDATIONS

This study investigated the effect of TETFUND interventions on research and development (R&D) in Nigerian tertiary institutions. Drawing on data from 160 academic staff across universities, polytechnics, and colleges of education, the findings reveal that research grant allocations and infrastructure availability significantly and positively influence research output. In contrast, participation in TETFUND-funded training did not show a significant effect on measurable research productivity within the period under review. The combined model explained approximately 21% of the variance in research output, confirming the overall importance of TETFUND interventions in enhancing R&D, though with varying degrees of effectiveness. These results suggest that while all three components of TETFUND are relevant, research funding and infrastructure provision have more immediate and measurable impacts on scholarly productivity than training programs alone.

This study recommends that:

- i. TETFUND and institutional administrators should ensure that research grant application processes are transparent, timely, and inclusive to allow broader participation among academic staff.

- ii. Given its significant influence on research output, TETFUND should continue investing in laboratory facilities, ICT centers, and research libraries, while also promoting regular maintenance to sustain long-term impact.
- iii. Training initiatives should be more research-focused, outcome-oriented, and tailored to faculty needs. Post-training monitoring and evaluation mechanisms should also be instituted to track their actual impact on research engagement and productivity.

This study contributes to both academic and policy literature in several ways. First, it provides empirical validation of the differential effects of TETFUND interventions on research and development outcomes, offering a nuanced understanding of what works and to what extent. Second, the study employs a multivariate analytical framework that simultaneously considers funding, training, and infrastructure, providing a more integrated and robust assessment than many earlier studies that examined each factor in isolation. Third, by using actual performance data (i.e., publication counts and project completions) rather than perceptions alone, the study adds methodological rigor to ongoing debates about the effectiveness of public funding in higher education. Lastly, the findings offer actionable insights to policymakers, educational planners, and university management teams aiming to optimize the design and implementation of research support programs in Nigeria's tertiary education sector. The study thus strengthens the empirical foundation for policy reforms that can drive innovation and academic excellence across the country.

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Appendix

```
. summarize rop rga ptt inf
```

Variable	Obs	Mean	Std. Dev.	Min	Max
rop	160	7.0625	2.836615	1	14
rga	160	3.1	1.501362	0	5
ptt	160	4.6125	3.252454	0	10
inf	160	1.6625	.800059	1	3

```
. swilk rop rga ptt inf
```

Shapiro-Wilk W test for normal data

Variable	Obs	W	V	z	Prob>z
rop	160	0.99454	0.672	-0.904	0.81705
rga	160	0.98401	1.967	1.538	0.06197
ptt	160	0.97886	2.600	2.173	0.01488
inf	160	0.98450	1.906	1.468	0.07111

```
. correlate rop rga ptt inf
(obs=160)
```

	rop	rga	ptt	inf
rop	1.0000			
rga	0.4253	1.0000		
ptt	0.0579	0.0325	1.0000	
inf	0.1867	0.0387	0.1355	1.0000

```
. regress rop rga ptt inf
```

Source	SS	df	MS	Number of obs =	160
Model	269.145013	3	89.7150044	F(3, 156) =	13.85
Residual	1010.22999	156	6.47583325	Prob > F =	0.0000
				R-squared =	0.2104
				Adj R-squared =	0.1952
Total	1279.375	159	8.04638365	Root MSE =	2.5448

rop	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
rga	.7899799	.1345718	5.87	0.000	.5241619	1.055798
ptt	.018816	.062651	0.30	0.764	-.1049377	.1425697
inf	.5941935	.2547502	2.33	0.021	.0909886	1.097398
_cons	3.538927	.6523764	5.42	0.000	2.250296	4.827558

```
. vif
```

Variable	VIF	1/VIF
inf	1.02	0.980450
ptt	1.02	0.980890
rga	1.00	0.997745
Mean VIF	1.01	

```
. hetttest
```

Breusch-Pagan / Cook-Weisberg test for heteroskedasticity

Ho: Constant variance

Variables: fitted values of rop

chi2(1) = 5.32

Prob > chi2 = 0.0211

```
. regress rop rga ptt inf, ro
```

Linear regression

Number of obs = 160

F(3, 156) = 13.06

Prob > F = 0.0000

R-squared = 0.2104

Root MSE = 2.5448

rop	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
rga	.7899799	.1450829	5.45	0.000	.5033995	1.07656
ptt	.018816	.0667653	0.28	0.778	-.1130646	.1506966
inf	.5941935	.2613951	2.27	0.024	.0778629	1.110524
_cons	3.538927	.6135449	5.77	0.000	2.326999	4.750854